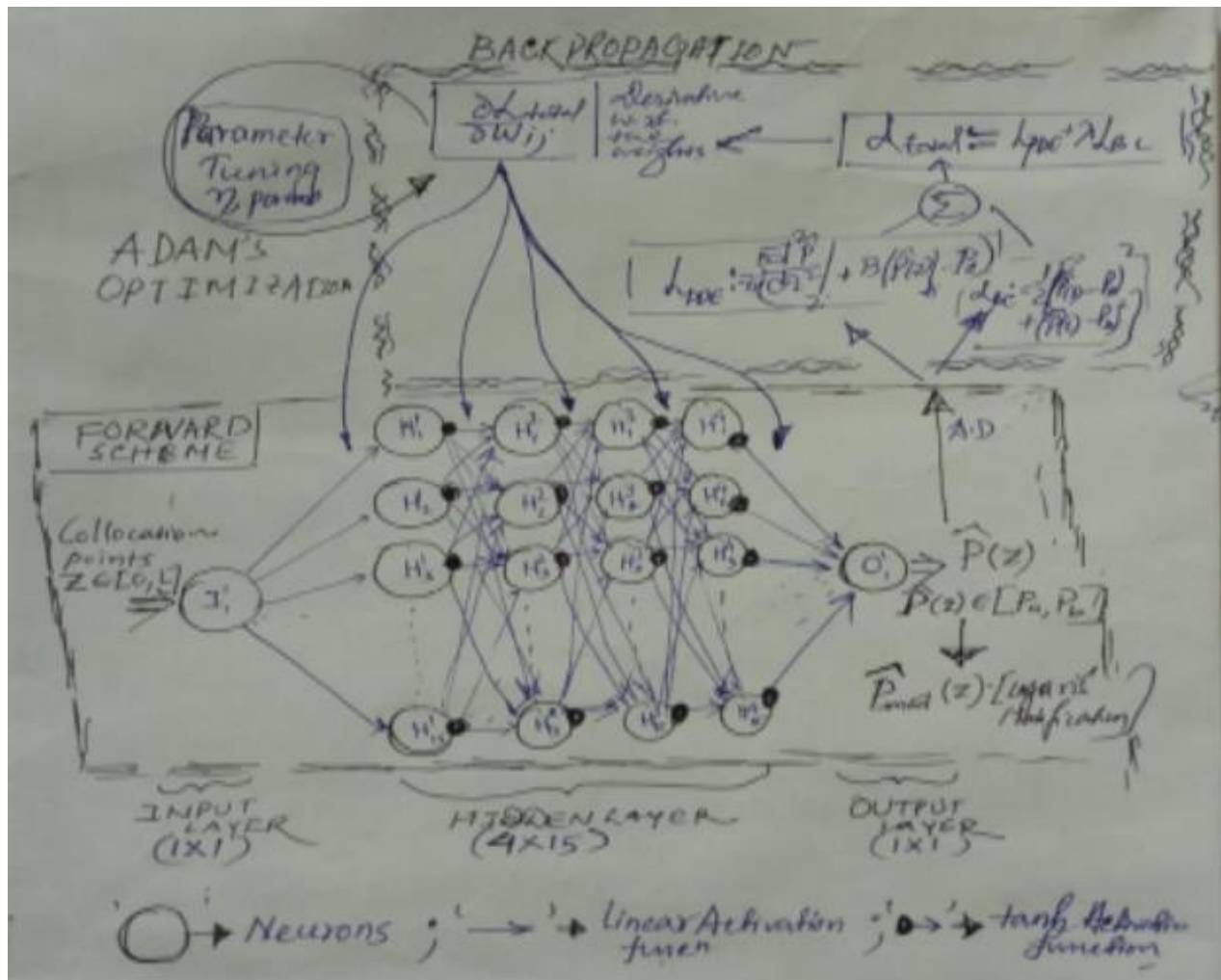


CREATIVE PROJECT ON “PHYSICS INFORMED NEURAL NETWORK FOR ELASTIC MODULUS IDENTIFICATION USING THE PRESSURE MEASUREMENT OF THE BLOOD” BY Souvik Sarkar (22ME10083) for the course on Microfluidics (ME60310)

1. This idea comes from the one of the Assignment which is given in the class about how the Diseases changes the Multiple properties (such as Stiffness, Compliance, Hydraulic Resistance or Impedence) and Inner and external part of blood vessel mechanics or the Micro-channels.
2. If we take the Pressure reading at some points of the vessel along the length of blood vessels . Then those measurement data can be directly fed into a Neural Network which will use the back-propagation technique to estimate the weights and finally estimate the Modulus of elasticity
3. Now this is the property which can be correlated with the Diseases , because diseases have different types of stiffness reduction. Create any Data science model which will find the correct diseases from the estimated modulus of the elasticity
4. Overall schematic of the structure in short way



5. Now the Loss function can be created as the PDE loss and Boundary condition losses both are weighted according to the needs where 'P(z)' is function of 'z'

PDE Loss is defined as :

$$\mathcal{L}_{PDE} = \frac{1}{n_c} \sum_{i=1}^{n_c} \left[\frac{d^2 \hat{P}}{dz^2} \Big|_{z_i} + B(\hat{P}(z_i) - P_c) \right]^2$$

B is scaling constant and Z_i is the collocation points.

BC loss :

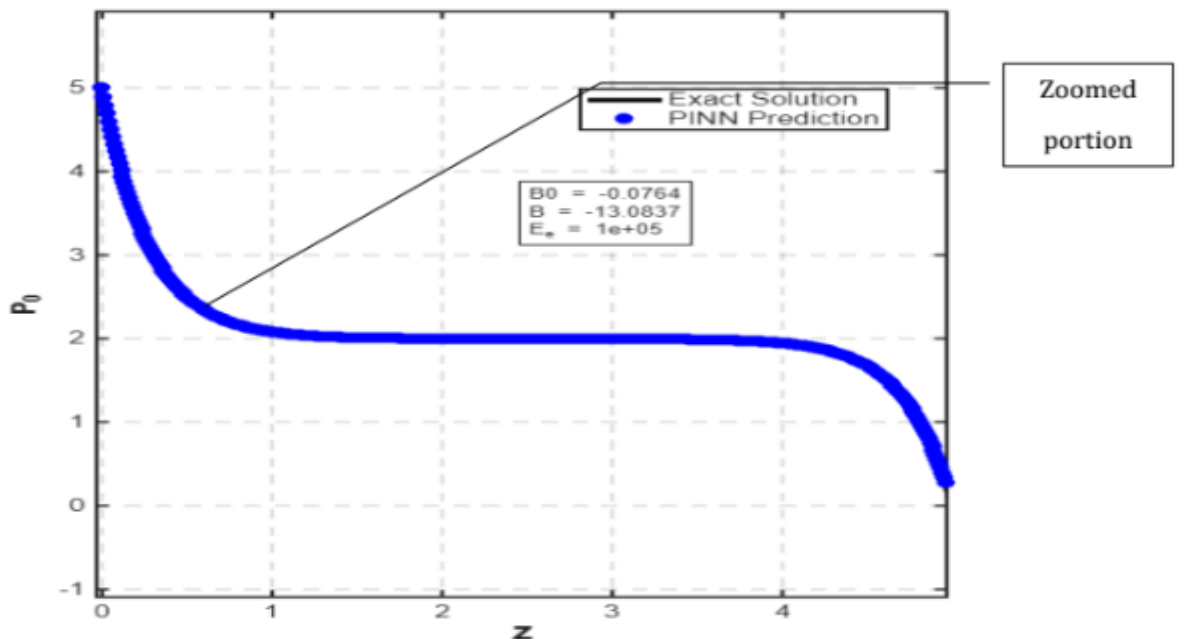
$$\mathcal{L}_{BC} = \frac{1}{2} \left[(\hat{P}(0) - P_a)^2 + (\hat{P}(L) - P_b)^2 \right]$$

Total Loss :

$$\mathcal{L}_{total} = \mathcal{L}_{PDE} + \lambda_{BC} \cdot \mathcal{L}_{BC}$$

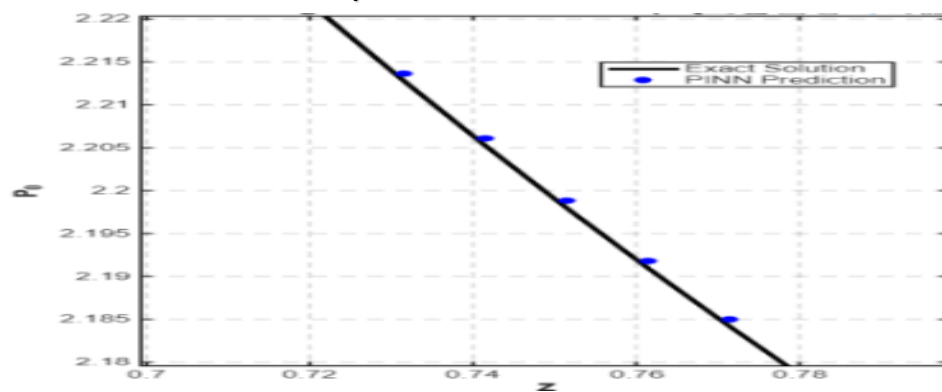
Where lambda is the weighting parameters

6. After taking some randomly arranged collocation points and use the NN feed forward mode to get the pressure
From the measurement of the pressure and the guessed value the NN will estimate the Modulus of elasticity for finding out the diseases
7. Forward Propagation Results :

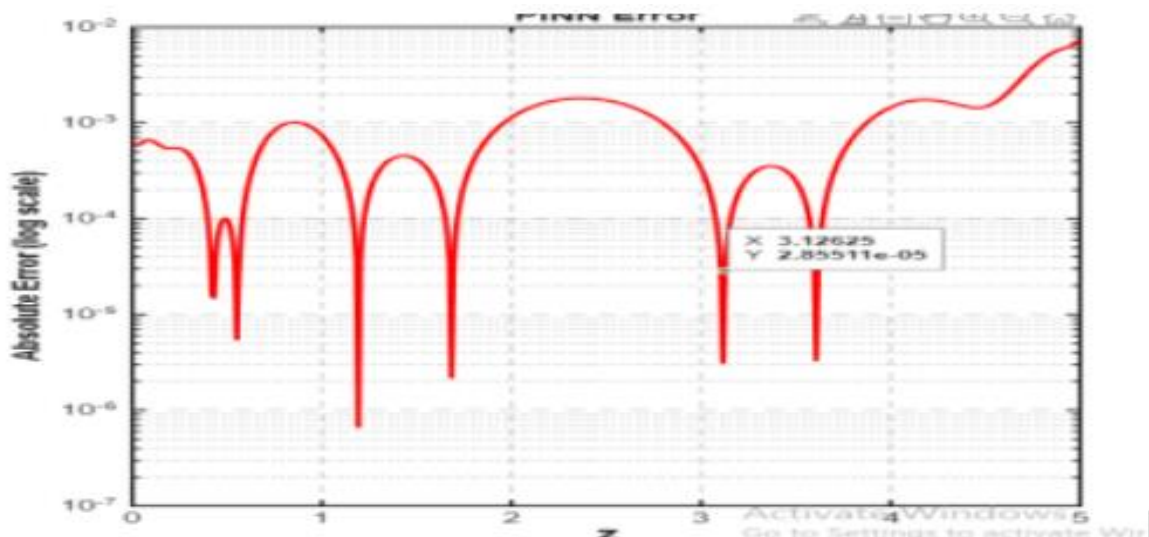


[P_0 (approximation in Lubrication theory) varying with 'z']

View of the Zoomed portion:



Total Losses over iteration :



**** N.B:** I have tried to check it in open source LLM model it says that all of these data points are like exactly following the analytical solution which is not good (Over-fitting) and These will results in the Curse of the Dimentionality . Also too many Neurons are chosen in each layer for just simple single pressure term which increase the cost of compuation. So I have to read and learn much more to work in this domain.

Ref: 'Peristaltic flow in the glymphatic system" (Romano et. Al 2020)

Code Link: https://github.com/sank-55/CSF_Flow_modelling-pinns- matlab-python/blob/main/PINN_PVS_flow/pinn_elastic_bc_pressuredist.M

REFERENCE PAPER Title : The attachment and detachment mechanism of silicone oil droplet in reversed electrowetting system on coplanar ITO electrode substrate by (2026) by Jian Zhang and Yinchuan Feng clarifies how frequency dictates whether a droplet sticks or flies off

When the AC frequency is lower than 800 Hz, the droplet always adheres to the substrate no matter how high the driving voltage is. As the AC frequency varies in the range of 850 Hz ~ 950 Hz, the droplet motion enters into the uncertain state. In this state, the droplet attaches or escapes mainly depending on whether the transient contact angle is larger than the critical contact angle, i.e., the critical contact angle of the droplet is the dominant factor determining the droplet detachment or attachment. When the AC frequency increases above 1 kHz, the droplet detaches from the substrate once the driving voltage is sufficiently high. Pictures form the paper :

Fig. 3 Evolution of the droplet profile at 0.5 kHz: s1 - s5 simulated by COMSOL; e1 - e5 captured by CCD

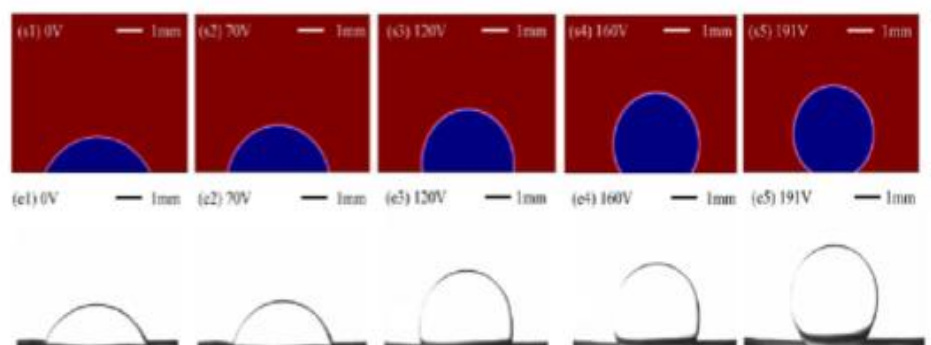
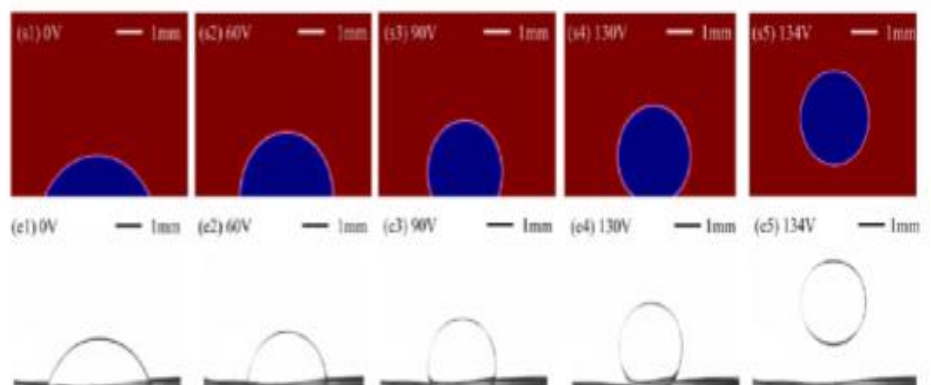


Fig. 7 Evolution of the droplet profile with voltage at 2 kHz: s1 - s5 simulated by COMSOL. e1 - e5 captured by CCD



[Droplet Contact angle with Varying AC Voltages]

Thanking You!